

ArithmeticSite

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Chapter 1

The multiplicative monoid $\mathbb{N}^\times$

Definition 1.1 (label=def:Nplus). Let $\mathbb{N}^\times$ denote the set $\{1, 2, 3, \dots\}$ of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

Definition 1.2 (label=def:BNplus, uses=def:Nplus). The *classifying category* $B\mathbb{N}^\times$ of $\mathbb{N}^\times$ is the one-object category whose single object is $*$ and whose morphisms $* \rightarrow *$ are the elements of $\mathbb{N}^\times$, with composition given by multiplication.

Chapter 2

The tropical semiring $\bar{\mathbb{N}}$

Definition 2.1 (label=def:Nbar). The *tropical semiring* $\bar{\mathbb{N}}$ is the set $\mathbb{N} \cup \{+\infty\}$ equipped with:

- *Tropical addition*: $a \oplus b := \min(a, b)$, with identity element $+\infty$.
- *Tropical multiplication*: $a \odot b := a + b$ (ordinary addition), with identity element 0.

In Lean, $\bar{\mathbb{N}}$ is represented as `Tropical (WithTop )`, via the abbreviation `NBar`. The zero element is `trop` (i.e. $+\infty$) and the multiplicative identity is `trop 0`.

Lemma 2.2 (label=lem:Nbar_semiring, uses = def : Nbar). $\bar{\mathbb{N}}$ is a commutative semiring.

Proof. This is the Mathlib instance `CommSemiring (Tropical (WithTop ))`, found by `inferInstance`. $\square$

Lemma 2.3 (label=lem:Nbar_idempotent, uses = def : Nbar). *Tropical addition on $\bar{\mathbb{N}}$ is idempotent: $a \oplus a = a$ for all $a \in \bar{\mathbb{N}}$.*

Proof. $a \oplus a = \min(a, a) = a$. In Lean this follows from `Tropical.add_eq_left`. $\square$

Definition 2.4 (label=def:frobenius, uses=def:Nbar). For each $n \in \mathbb{N}^\times$, the *Frobenius endomorphism* $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is defined by

$$\varphi_n(x) := n + x \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where $+$ denotes ordinary addition in $\mathbb{N} \cup \{+\infty\}$. Equivalently, in `Tropical (WithTop )`, this is tropical multiplication by `trop (n : WithTop )`, i.e. $\varphi_n(x) = \text{trop } n \odot x$.

In Lean, φ_n is defined as an `AddMonoid.End NBar` via `frobeniusEndomorphism`.

Lemma 2.5 (label=lem:frobenius_addmon_hom, uses = def : frobenius). For each $n \in \mathbb{N}^\times$, the map $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is an additive monoid endomorphism, i.e. it preserves tropical addition and the zero element $+\infty$:

$$\begin{aligned} \varphi_n(+\infty) &= +\infty, \\ \varphi_n(a \oplus b) &= \varphi_n(a) \oplus \varphi_n(b). \end{aligned}$$

The second identity follows from the distributivity $n + \min(a, b) = \min(n + a, n + b)$.

Remark 2.6. Note that φ_n is *not* a semiring homomorphism in general, since it does not preserve the multiplicative identity: $\varphi_n(0) = n \neq 0$ for $n > 0$. It is however a homomorphism of $\bar{\mathbb{N}}$ -modules.

Lemma 2.7 (label=lem:frobenius_comp, uses = def : frobenius). *The Frobenius endomorphisms satisfy the composition law*

$$\varphi_m \circ \varphi_n = \varphi_{m+n}$$

for all $m, n \in \mathbb{N}^\times$. In particular, the map $n \mapsto \varphi_n$ is a semigroup homomorphism from $(\mathbb{N}^\times, +)$ into the endomorphism semigroup $\text{End}_{\text{AddMon}}(\mathbb{N})$.

Proof. For any $x \in \bar{\mathbb{N}}$: $(\varphi_m \circ \varphi_n)(x) = m + (n + x) = (m + n) + x = \varphi_{m+n}(x)$. In Lean this is `frobenius_comp`, proved via `mul_assoc` in `Tropical (WithTop)`. $\square$

Chapter 3

The presheaf topos $\widehat{\mathbb{N}^\times}$

Definition 3.1 (label=def:presheaf_topos, uses = def : BNplus). The *presheaf topos* $\widehat{\mathbb{N}^\times}$ is the `caNplus` is the category of functors $\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$, i.e. the category of sets equipped with a left $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative N+)`.

Lemma 3.2 (label=lem:presheaf_topos_is_topos, uses = def : presheaf_topos). $\widehat{\mathbb{N}^\times}$ is a Grothendieck topos.

Proof. Any presheaf category on a small category is a Grothendieck topos. □

Chapter 4

The structure sheaf

Definition 4.1 (label=def:structure_ssheaf, uses = def : presheaf_topos, def : Nbar, def : frobenius). The structure sheaf $\mathcal{O}$ of the Arithmetic Site is the object of $\widehat{\mathbb{N}^\times}$ whose underlying set is $\bar{\mathbb{N}}$ and whose $\mathbb{N}^\times$ -action is given by the Frobenius endomorphisms: $n \cdot x := \varphi_n(x)$.

Lemma 4.2 (label=lem:structure_ssheaf_action, uses = def : structure_ssheaf, lem : frobenius_comp). The assignment $n \mapsto \varphi_n$ defines a semigroup homomorphism $(\mathbb{N}^\times, +) \rightarrow \text{End}_{\text{AddMon}}(\bar{\mathbb{N}})$, making $\bar{\mathbb{N}}$ into a left $(\mathbb{N}^\times, +)$ -module.

Chapter 5

The Arithmetic Site

Definition 5.1 (label=def:arithmetic_site, uses = def : presheaf_topos, def : structure_sheaf). The *Arithmetic Site* is the pair $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ consisting of the presheaf topos $\widehat{\mathbb{N}^\times}$ together with its structure sheaf $\mathcal{O}$.

Chapter 6

Points of the topos

Definition 6.1 (label=def:topos_point, uses = def : presheaf_topos). A *point* of a topos $\mathcal{E}$ is a geometric morphism $p : \mathbf{Set} \rightarrow \mathcal{E}$, i.e. an adjoint pair $p^* \dashv p_*$ where p^* is left exact.

Theorem 6.2 (label=thm:points_are_adelic_classes, uses = def : topos_point, def : arithmetic_site). The points of $\widehat{\mathbb{N}^\times}$ are in natural bijection with the adèle class space $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$.

Proof.

□

Chapter 7

The Frobenius correspondences

Definition 7.1 (label=def:frobenius_ccorrespondence, uses = def : arithmetic_site). For each $\lambda \in \mathbb{Q}_+^\times$, the *Frobenius correspondence* Φ_λ is a correspondence on $\widehat{\mathbb{N}}^\times \times \widehat{\mathbb{N}}^\times$ arising from the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space.

Theorem 7.2 (label=thm:frobenius_are_ccorrespondences, uses = def : frobenius_ccorrespondence, thm : points_are_ade). *Frobenius correspondences realize the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space at the level of the topos $\widehat{\mathbb{N}}^\times$.*

Proof.

□

Chapter 8

Connection to the Riemann zeta function

Theorem 8.1 (label=thm:zeta_counting,uses = thm : frobenius_correspondences). *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function $\zeta(s)$.*

Proof.

□